

Exercise Sheet - Python

Exercise 1

This exercise is very helpful to revise lists, loops, functions

1. Ask the user for their favorite fruits and print them
2. Using a while loop, ask the user for the operation below
3. Keep asking the user for the operation until he says no
4. Do a function for each operation example:

```
def addLast(list, val):  
    list.append(val)
```
5. Use match to choose the correct function (the user enter numbers based on your text)
6. Print the list after each operation
7. If the user said no then print Bye

Operations needed:

1. AddLast: Add a value at the end
2. AddAt: Add at a Specific Position
3. AddMultiple: Add multiple items
4. RemoveItem: remove the value at first occurrence
5. RemoveLast: remove last item
6. RemoveAt: remove at specific position
7. SortList: sort the list
8. ReverseList: reverse the list
9. CountValue: count a Specific value
10. FindItem: find index of first occurrence of an item
11. CheckItem: check if an item in the list

Exercise 2

Vocabulary App

Using dictionary, loops, conditions

1. Do a program that have the gives the user some options on a vocabulary dictionary to *add a word, look up a word, remove a word, show all words*
2. Keep asking until the user want to exit

Exercise 3 - Advanced level

You are asked to build a library management program that keeps track of books and their borrowing status.

The program should store multiple books, and for each book, store detailed information.

Data Organization

Choose a suitable dictionary structure where:

- each book can be accessed using its title
- each book stores:
 - the author's name
 - how many copies exist
 - which users have borrowed it

You must decide how this structure will look in Python.

Program Flow

1. Display a list of operations to the user.
2. Ask the user to choose an operation using a number.
3. Perform the selected operation.
4. Repeat the process until the user chooses to exit.
5. After each operation, show the current state of the library.
6. When the program ends, display a goodbye message.

Operations Needed

1. AddBook: Add a new book with author and number of copies
2. RemoveBook: Remove a book completely
3. BorrowBook: Add a borrower if copies are available
4. ReturnBook: Remove a borrower from the book
5. AddCopies: Increase the number of copies
6. RemoveCopies: Decrease the number of copies (cannot be less than borrowed books)
7. ShowBookInfo: Print all information about a specific book
8. ShowAllBooks: Print all books in the library
9. CountBorrowers: Count how many borrowers a book has
10. SearchByAuthor: Show all books written by a specific author
11. Exit

Constraints & Considerations

A book cannot be borrowed if no copies are available.

The number of copies cannot be reduced below the number of borrowers.

All checks (book existence, valid operations) must be handled safely.